

Client VM Side (Theory Section)

"The Ubuntu client VM initially failed to obtain DHCP from the OPNsense LAN due to a VMware VMnet mismatch (adapter was on VMnet0/bridged rather than VMnet1/host-only). After correcting the VMnet assignment, the client still retained its previous NAT IP (192.168.169.132) alongside the new LAN IP. A static configuration of 10.10.10.150/24 with gateway 10.10.10.1 and DNS 10.10.10.1 was applied via nmcli to establish the correct LAN path.

An additional DNS issue was resolved by changing the NetworkManager DNS configuration from the stale NAT DNS (192.168.169.1) to the firewall LAN DNS (10.10.10.1) and setting ignore-auto-dns.

A subtle but critical firewall configuration issue was identified: the LAB2 block rules initially used Source = 'LAN address' (the firewall's own IP), which did not match client traffic. Changing Source to 'LAN net' (10.10.10.0/24) caused the rules to correctly match and block the intended traffic."

Completion Questions (Lab Page 6)

38. Why must the specific block rules be placed above the broad allow rule?

"OPNsense evaluates interface rules from top to bottom, and the first matching rule determines the action (first-match-wins with 'quick' flag). If the broad Default allow rule were above the specific block rules, it would match the traffic first and permit it, preventing the block rules from ever being reached."

39. Which five packet attributes are most useful when explaining a firewall decision?

"Interface, source IP, destination IP, protocol, and destination port."

40. Why did blocking ICMP not block HTTPS?

"The block rule specified Protocol = ICMP. HTTPS uses TCP on port 443, so its packets did not match the ICMP rule. This demonstrates protocol-specific filtering."

41. What difference did you observe between blocked TCP port 80 traffic and permitted TCP port 443 traffic?

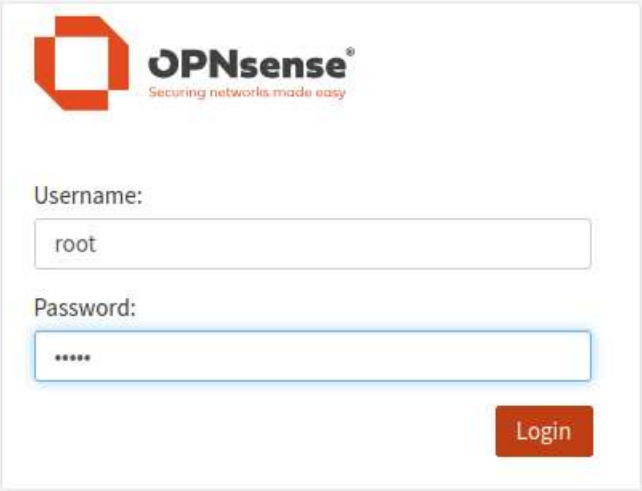
"Blocked port-80 traffic showed only repeated TCP SYN packets with no SYN-ACK response (silent drop). Permitted port-443 traffic showed a complete three-way handshake (SYN, SYN-ACK, ACK) followed by encrypted TLS data exchange."

42. What role does outbound NAT play when icdfa-nslab-client-v1 uses a private IPv4 address?

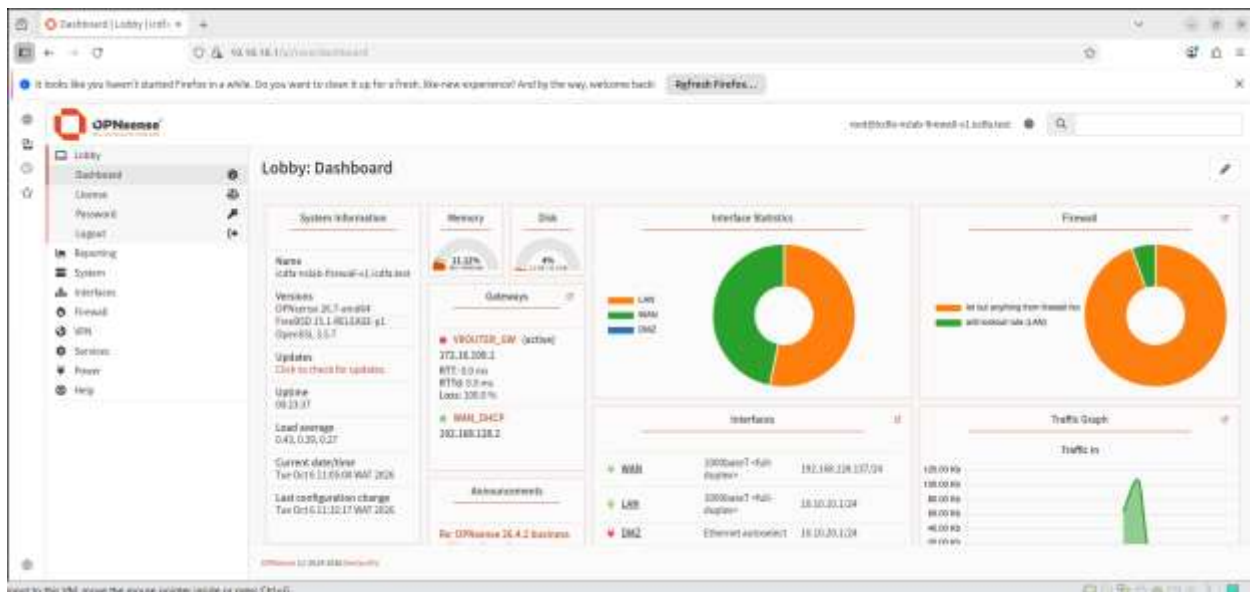
"Outbound NAT translates the client's private source IP (10.10.10.150) to the firewall's WAN IP (192.168.128.137) before packets leave the WAN interface, allowing the private LAN to reach the public internet. The state table preserves the mapping so responses are translated back to the original client."

43. Why is restoring the original state an important part of a controlled security laboratory?

"Restoring the baseline ensures the environment is in a known-good state for the next activity, prevents leftover block rules from causing unexpected failures, and allows subsequent experiments to start from a consistent starting point for accurate comparison."



The image shows the OPNsense login interface. At the top left is the OPNsense logo, which consists of an orange square with a white 'O' inside, followed by the text 'OPNsense®' and the tagline 'Securing networks made easy' in a smaller font. Below the logo, there are two input fields: 'Username:' with the text 'root' entered, and 'Password:' with six asterisks '*****' entered. To the right of the password field is a red 'Login' button. At the bottom of the interface, there is a copyright notice: 'OPNsense (c) 2014-2026 Deciso B.V.'.



The screenshot shows the OPNsense Firewall Rule configuration window titled "Edit rule". The window is in "advanced mode" and has a "full help" link. The configuration is organized into sections:

- Organisation:**
 - Enabled:** ☒
 - Categories:** Nothing selected (Clear All, Select All)
 - Description:** LAB2 BLOCK ICMP TO 1.1.1.1
- Interface:**
 - Invert Interface:** ☐
 - Interface:** LAN (Clear All, Select All)
- Filter:**
 - Quick:** ☒

At the bottom of the window are "Cancel" and "Save" buttons.

Edit rule



i Interface	LAN
x Clear All ✓ Select All	
▼ Filter	
i Quick	☒
i Action	Block
i Direction	In
i Version	IPv4
i Protocol	ICMP
i ICMP type	Nothing selected
x Clear All ✓ Select All	
i Invert Source	☐

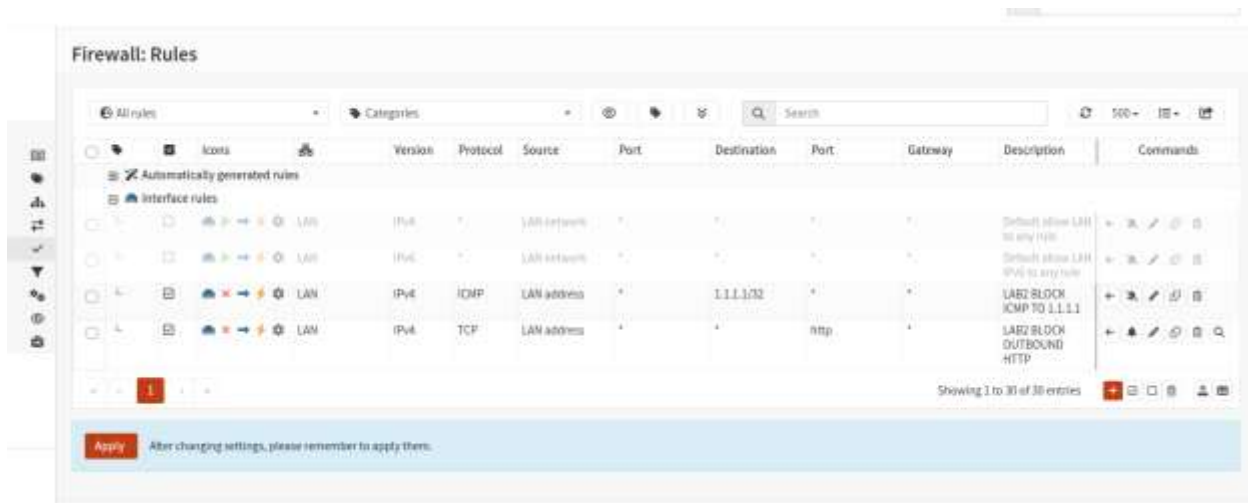
CancelSave

Edit rule

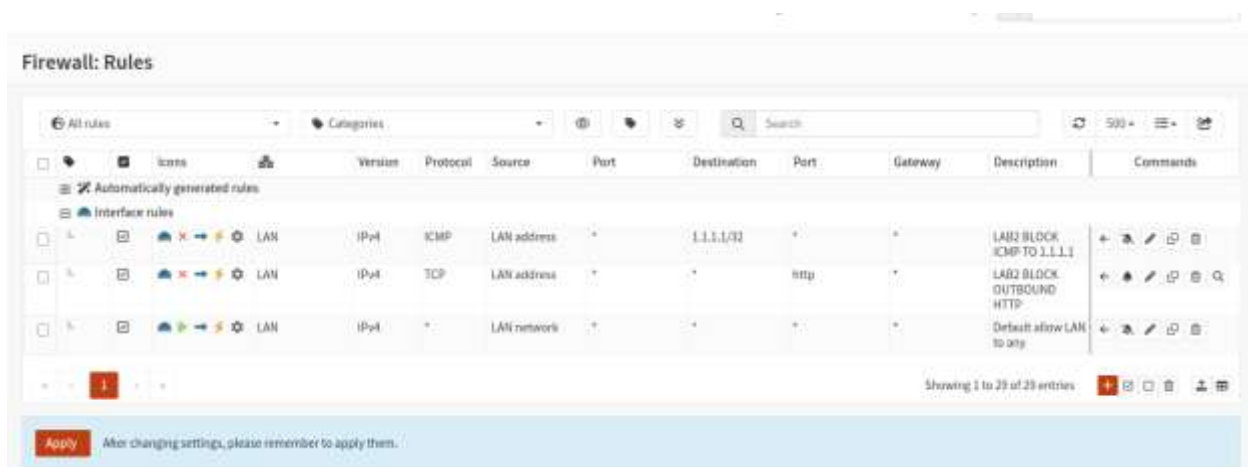
i Invert Source	☐
i Source	LAN address
i Source Port	any
i Invert Destination	☐
i Destination	Single host or Network
	1.1.1.1/32
i Destination Port	any
i Log	☐
▼ Source Routing	
i Gateway	None

 Version	IPv4
 Protocol	TCP
 Invert Source	☐
 Source	LAN address
 Source Port	any
 Invert Destination	☐
 Destination	any
 Destination Port	HTTP (80)
 Log	☒
▼ Source Routing	
 Gateway	None

▼ Organisation	
 Enabled	☒
 Categories	Nothing selected  Clear All  Select All
 Description	LAB2 BLOCK OUTBOUND HTTP
▼ Interface	
 Invert Interface	☐
 Interface	LAN  Clear All  Select All
▼ Filter	
 Quick	☒
 Action	Block



"LAN rules page showing both LAB2 block rules positioned above the Default allow LAN to any rule, as required for top-down rule processing. Both rules are enabled with logging. The Default allow rule was temporarily disabled during rule creation to ensure the block rules were positioned first, then re-enabled."



"LAN rules page showing both LAB2 block rules (LAB2 BLOCK ICMP TO 1.1.1.1 and LAB2 BLOCK OUTBOUND HTTP) positioned above the Default allow LAN to any rule, ensuring top-down firewall evaluation blocks the specified traffic before the broad allow rule is reached."

```
samina@samina-VMware-Virtual-Platform:~$ ping -c 4 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data.

--- 1.1.1.1 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3065ms

samina@samina-VMware-Virtual-Platform:~$
```

```

192.168.1.10/24 dev enp5s3 proto kernel scope link src 192.168.1.10
samina@samina-VMware-Virtual-Platform:~$ ping -c 4 10.10.10.1
getent hosts example.com
curl -I https://example.com
PING 10.10.10.1 (10.10.10.1) 56(84) bytes of data.
64 bytes from 10.10.10.1: icmp_seq=1 ttl=64 time=0.614 ms
64 bytes from 10.10.10.1: icmp_seq=2 ttl=64 time=2.17 ms
64 bytes from 10.10.10.1: icmp_seq=3 ttl=64 time=2.77 ms
64 bytes from 10.10.10.1: icmp_seq=4 ttl=64 time=0.487 ms

--- 10.10.10.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3064ms
rtt min/avg/max/mdev = 0.487/1.511/2.773/0.984 ms
2606:4700:10::6814:179a example.com
2606:4700:10::ac42:93f3 example.com
HTTP/2 200
date: Tue, 06 Oct 2026 11:06:17 GMT
content-type: text/html; charset=utf-8
server: cloudflare
last-modified: Fri, 02 Oct 2026 16:11:02 GMT
allow: GET, HEAD
accept-ranges: bytes
age: 10449
cf-cache-status: HIT
cf-ray: a4643221cdd152a1-KHI
alt-svc: h3=":443"; ma=86400

samina@samina-VMware-Virtual-Platform:~$

```

```

samina@samina-VMware-Virtual-Platform:~$ ping -c 4 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data.

--- 1.1.1.1 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3055ms

```

“After applying the LAB2 BLOCK ICMP rule: ping to 1.1.1.1 (external DNS) fails with 100% packet loss, while ping to the LAN gateway (10.10.10.1), DNS resolution for example.com, and HTTPS access to example.com (HTTP/2 200 via Cloudflare) remain fully functional demonstrating protocol-specific filtering rather than loss of internet connectivity.”

```

samina@samina-VMware-Virtual-Platform:~$ curl --max-time 10 -I http://example.com
curl --max-time 10 -I https://example.com
HTTP/1.1 200 OK
Date: Tue, 06 Oct 2026 11:10:11 GMT
Content-Type: text/html; charset=utf-8
Connection: keep-alive
Server: cloudflare
Last-Modified: Sun, 04 Oct 2026 20:44:03 GMT
Allow: GET, HEAD
Accept-Ranges: bytes
Age: 10692
cf-cache-status: HIT
CF-RAY: a46437d6fc6ac972-KHI
alt-svc: h3=":443"; ma=86400

HTTP/2 200
date: Tue, 06 Oct 2026 11:10:11 GMT
content-type: text/html; charset=utf-8
server: cloudflare
last-modified: Fri, 02 Oct 2026 16:11:02 GMT
allow: GET, HEAD
accept-ranges: bytes
age: 10682
cf-cache-status: HIT
cf-ray: a46437d7e81552a1-KHI
alt-svc: h3=":443"; ma=86400

```

"After applying the LAB2 BLOCK OUTBOUND HTTP rule: HTTP request to example.com (TCP port 80) times out, while the HTTPS request (TCP port 443) returns HTTP/2 200 — confirming port-specific blocking on TCP/80 with TCP/443 fully permitted."

The screenshot shows the OPNsense Firewall Log Files: Live View interface. The table displays log entries for blocked traffic. The columns are Interface, Time, Protocol, Source, Destination, Action, and Label. The entries show that traffic from 192.168.128.127 to 1.1.1.1 on port 80 is blocked by the rule 'let out anything from firewall host itself (force pol)'.

Interface	Time	Protocol	Source	Destination	Action	Label
WAN	2026-10-06T11:40:14	HTTP	192.168.128.127	1.1.1.1	pass	let out anything from firewall host itself (force pol)
WAN	2026-10-06T11:40:12	HTTP	192.168.128.127	1.1.1.1	pass	let out anything from firewall host itself (force pol)
WAN	2026-10-06T11:40:10	HTTP	192.168.128.127	1.1.1.1	pass	let out anything from firewall host itself (force pol)
WAN	2026-10-06T11:39:58	HTTP	192.168.128.127:4284	1.1.1.1:4443	pass	let out anything from firewall host itself (force pol)
WAN	2026-10-06T11:39:55	HTTP	192.168.128.127:3291	1.1.1.1:4443	pass	let out anything from firewall host itself (force pol)
WAN	2026-10-06T11:39:54	HTTP	192.168.128.127:3206	1.1.1.1:4443	pass	let out anything from firewall host itself (force pol)
WAN	2026-10-06T11:39:53	HTTP	192.168.128.127:3041	1.1.1.1:4443	pass	let out anything from firewall host itself (force pol)
WAN	2026-10-06T11:39:51	HTTP	192.168.128.127:3758	1.1.1.1:4443	pass	let out anything from firewall host itself (force pol)
WAN	2026-10-06T11:39:51	HTTP	192.168.128.127:11476	1.1.1.1:4443	pass	let out anything from firewall host itself (force pol)

Firewall: Log Files: Live View

Filters

Quick search (all fields)...

action

contains

Search...

Apply

Templates

»

Choose template

+

Options

☒

Auto refresh

☐

Lookup hostnames

☐

Select any of given criteria (or)

25

+

Table size

10000

+

History size

Active filters:

Interface	Time	Protocol	Source	Destination	Action	Label		
WAN	Out	2026-10-06T12:17:15	TCP	192.168.128.137:23816	104.20.23.154:80	pass	let out anything from firewall host itself (force gw)	🔍
WAN	Out	2026-10-06T12:17:15	UDP	192.168.128.137:18855	172.64.32.162:53	pass	let out anything from firewall host itself (force gw)	🔍
LAN	In	2026-10-06T12:17:09	TCP	10.10.10.150:39380	10.10.10.1:443	pass	anti-knockout rule	🔍
LAN	In	2026-10-06T12:17:05	ICMP	10.10.10.150	1.1.1.1	block	LAB2 BLOCK ICMP TO 1.1.1.1	🔍
LAN	In	2026-10-06T12:17:04	ICMP	10.10.10.150	1.1.1.1	block	LAB2 BLOCK ICMP TO 1.1.1.1	🔍
LAN	In	2026-10-06T12:17:03	ICMP	10.10.10.150	1.1.1.1	block	LAB2 BLOCK ICMP TO 1.1.1.1	🔍
LAN	In	2026-10-06T12:17:02	ICMP	10.10.10.150	1.1.1.1	block	LAB2 BLOCK ICMP TO 1.1.1.1	🔍
WAN	Out	2026-10-06T12:16:50	UDP	192.168.128.137:133	203.86.62.214:123	pass	let out anything from firewall host itself (force gw)	🔍
WAN	Out	2026-10-06T12:16:45	UDP	192.168.128.137:133	103.367.339.34:123	pass	let out anything from firewall host itself (force gw)	🔍

OPNsense [v21.01.0-RELEASE] [v21.01.0-RELEASE]

OPNsense Firewall → Log Files → Live View showing 4 blocked ICMP packets from Ubuntu client (10.10.10.150) to 1.1.1.1, each matched against the LAB2 BLOCK ICMP TO 1.1.1.1 rule with action=block. This confirms top-down rule processing: the specific block rule matched before the Default allow rule was reached."

10000 History size

Interface	Time	Protocol	Source	Destination	Action	Label		
WAN	Out	2026-10-06T12:19:43	UDP	192.168.128.137:123	203.86.62.214:123	pass	let out anything from firewall host itself (force gw)	🔴
LAN	In	2026-10-06T12:19:43	TCP	10.10.10.150:30656	104.20.23.154:80	block	LAB2 BLOCK OUTBOUND HTTP	🔴
LAN	In	2026-10-06T12:19:43	TCP	10.10.10.150:30656	104.20.23.154:80	block	LAB2 BLOCK OUTBOUND HTTP	🔴
LAN	In	2026-10-06T12:19:40	TCP	10.10.10.150:39556	104.20.23.154:80	block	LAB2 BLOCK OUTBOUND HTTP	🔴
LAN	In	2026-10-06T12:19:40	TCP	10.10.10.150:45628	172.64.147.243:80	block	LAB2 BLOCK OUTBOUND HTTP	🔴
WAN	Out	2026-10-06T12:19:40	UDP	192.168.128.137:123	157.20.147.14:123	pass	let out anything from firewall host itself (force gw)	🔴
LAN	In	2026-10-06T12:19:38	TCP	10.10.10.150:45628	172.64.147.243:80	block	LAB2 BLOCK OUTBOUND HTTP	🔴
LAN	In	2026-10-06T12:19:38	TCP	10.10.10.150:45628	172.64.147.243:80	block	LAB2 BLOCK OUTBOUND HTTP	🔴
WAN	Out	2026-10-06T12:19:28	UDP	192.168.128.137:13084	185.125.250.65:53	pass	let out anything from firewall host itself (force gw)	🔴

"OPNsense Firewall → Log Files → Live View showing blocked traffic matched against both LAB2 block rules: (1) ICMP echo requests from Ubuntu (10.10.10.150) to 1.1.1.1 blocked by 'LAB2 BLOCK ICMP TO 1.1.1.1'; (2) TCP SYN packets to port 80 blocked by 'LAB2 BLOCK OUTBOUND HTTP'. This confirms top-down rule processing

```
root@icdfa-ns-lab-firewall-v1:~ # tcpdump -i em1 -n icmp
tcpdump: verbose output suppressed, use -v[-v]... for full protocol decode
listening on em1, link-type EN10MB (Ethernet), snapshot length 262144 bytes
12:21:00.643045 IP 10.10.10.150 > 1.1.1.1: ICMP echo request, id 8159, seq 1, length 64
12:21:01.662681 IP 10.10.10.150 > 1.1.1.1: ICMP echo request, id 8159, seq 2, length 64
12:21:02.687212 IP 10.10.10.150 > 1.1.1.1: ICMP echo request, id 8159, seq 3, length 64
12:21:03.711840 IP 10.10.10.150 > 1.1.1.1: ICMP echo request, id 8159, seq 4, length 64
```

"tcpdump on OPNsense LAN interface (em1): 4 ICMP echo requests from Ubuntu (10.10.10.150) to 1.1.1.1, with no echo replies visible

confirming the LAB2 BLOCK ICMP rule dropped the packets before they reached the internet." the specific block rules matched before the Default allow rule could apply."

```
12:22:57.296545 IP 10.10.10.150.37458 > 104.20.23.154.80: Flags [S], seq 4613240
0, win 64240, options [mss 1460,sackOK,TS val 2960053221 ecr 0,nop,wscale 9], le
ngth 0
12:22:57.616254 IP 10.10.10.150.38678 > 140.248.129.91.80: Flags [S], seq 408471
8887, win 64240, options [mss 1460,sackOK,TS val 3303451112 ecr 0,nop,wscale 9],
length 0
12:22:57.873180 IP 10.10.10.150.38694 > 140.248.129.91.80: Flags [S], seq 240200
7547, win 64240, options [mss 1460,sackOK,TS val 2134965636 ecr 0,nop,wscale 9],
length 0
12:22:58.321438 IP 10.10.10.150.37458 > 104.20.23.154.80: Flags [S], seq 4613240
0, win 64240, options [mss 1460,sackOK,TS val 2960054245 ecr 0,nop,wscale 9], le
ngth 0
12:22:58.640495 IP 10.10.10.150.38678 > 140.248.129.91.80: Flags [S], seq 408471
8887, win 64240, options [mss 1460,sackOK,TS val 3303452136 ecr 0,nop,wscale 9],
length 0
12:22:58.896500 IP 10.10.10.150.38694 > 140.248.129.91.80: Flags [S], seq 240200
7547, win 64240, options [mss 1460,sackOK,TS val 2134966660 ecr 0,nop,wscale 9],
length 0
12:22:59.665141 IP 10.10.10.150.38678 > 140.248.129.91.80: Flags [S], seq 408471
8887, win 64240, options [mss 1460,sackOK,TS val 3303453160 ecr 0,nop,wscale 9],
length 0
12:22:59.921224 IP 10.10.10.150.38694 > 140.248.129.91.80: Flags [S], seq 240200
7547, win 64240, options [mss 1460,sackOK,TS val 2134967684 ecr 0,nop,wscale 9],
length 0
```

"tcpdump on OPNsense LAN interface (em1): TCP SYN packets from Ubuntu (10.10.10.150) to multiple destinations on port 80, with no SYN-ACK responses and retransmissions confirming the LAB2 BLOCK OUTBOUND HTTP rule silently dropped the traffic."

```
gth 375
12:25:19.368918 IP 10.10.10.150.39380 > 10.10.10.1.443: Flags [I], ack 106483, w
in 1157, options [nop,nop,TS val 3014505469 ecr 774430270], length 0
12:25:19.369006 IP 10.10.10.1.443 > 10.10.10.150.39380: Flags [P], seq 106858:1
06889, ack 6470, win 514, options [nop,nop,TS val 774430270 ecr 3014505400], len
gth 31
12:25:19.369087 IP 10.10.10.150.39380 > 10.10.10.1.443: Flags [I], ack 106858, w
in 1157, options [nop,nop,TS val 3014505469 ecr 774430270], length 0
12:25:19.369473 IP 10.10.10.150.39380 > 10.10.10.1.443: Flags [I], ack 106889, w
in 1157, options [nop,nop,TS val 3014505469 ecr 774430270], length 0
12:25:20.457418 IP 10.10.10.150.39380 > 10.10.10.1.443: Flags [P], seq 6470:659
2, ack 106889, win 1157, options [nop,nop,TS val 3014506556 ecr 774430270], leng
th 122
12:25:20.457572 IP 10.10.10.1.443 > 10.10.10.150.39380: Flags [I], ack 6592, win
514, options [nop,nop,TS val 774431363 ecr 3014506556], length 0
12:25:20.528491 IP 10.10.10.1.443 > 10.10.10.150.39380: Flags [P], seq 106889:1
07853, ack 6592, win 514, options [nop,nop,TS val 774431430 ecr 3014506556], len
gth 964
12:25:20.533816 IP 10.10.10.150.39380 > 10.10.10.1.443: Flags [I], ack 107853, w
in 1157, options [nop,nop,TS val 3014506628 ecr 774431430], length 0
^C
427 packets captured
503 packets received by filter
0 packets dropped by kernel
root@icdf-nlab-firoull-ud1:~ #
```

"tcpdump on OPNsense LAN interface (em1): established TCP connection on port 443 with SYN/ACK, ACK, and PUSH (data) flags — showing a completed TLS handshake and

encrypted traffic. This confirms HTTPS traffic is permitted while port 80 and ICMP were blocked."

"tcpdump capture on OPNsense LAN interface (em1) verifying firewall decisions: (1) ICMP echo requests with no replies (blocked); (2) TCP SYN packets to port 80 without SYN-ACK (blocked); (3) successful TCP handshake and TLS traffic on port 443 (permitted)."

Firewall: Diagnostics: States

Select rule

int	Dir	Proto	Source	Nat	Destination	Gateway	State	Rule	Commands
all	←	icmp	192.168.128.137:47878		172.16.100.1:8		0:0	let out anything from fir...	
all	→	tcp	10.10.10.150:51010		34.107.243.93:443		ESTABLISHED:ESTABLIS...	Default allow LAN to any	
all	←	tcp	192.168.128.137:48622	10.10.10.150:51010	34.107.243.93:443		ESTABLISHED:ESTABLIS...	let out anything from fir...	
all	→	tcp	10.10.10.150:39390		10.10.10.1:443		FIN_WAIT_2:FIN_WAIT_2	anti-spoofing rule	
all	→	tcp	10.10.10.150:39870		10.10.10.1:53		FIN_WAIT_2:FIN_WAIT_2	Default allow LAN to any	
all	←	udp	192.168.128.137:123		10.1.1.1:123		MULTIPLE-SINGLE	let out anything from fir...	
all	←	udp	192.168.128.137:123		157.20.147.14:123		MULTIPLE-SINGLE	let out anything from fir...	
all	←	udp	192.168.128.137:123		153.167.139.26:123		MULTIPLE-SINGLE	let out anything from fir...	
all	←	udp	192.168.128.137:123		153.167.139.26:123		MULTIPLE-SINGLE	let out anything from fir...	

Showing 1 to 13 of 13 entries

"OPNsense Firewall → Diagnostics → States showing an active HTTPS session (Destination 34.107.243.93:443) with outbound NAT translation: the LAN-side state shows Source 10.10.10.150:51010 with Rule 'Default allow LAN to any'; the WAN-side state shows Source 192.168.128.137:48622 with Nat field 10.10.10.150:51010 confirming automatic outbound NAT is translating the private LAN source address for internet access."

```
curl --max-time 10 -I https://example.com
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data.
64 bytes from 1.1.1.1: icmp_seq=1 ttl=127 time=26.6 ms
64 bytes from 1.1.1.1: icmp_seq=2 ttl=127 time=26.1 ms
64 bytes from 1.1.1.1: icmp_seq=3 ttl=127 time=26.4 ms
64 bytes from 1.1.1.1: icmp_seq=4 ttl=127 time=27.2 ms

--- 1.1.1.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 26.135/26.558/27.166/0.381 ms
HTTP/1.1 200 OK
Date: Tue, 06 Oct 2026 11:28:53 GMT
Content-Type: text/html; charset=utf-8
Connection: keep-alive
Server: cloudflare
Last-Modified: Sun, 04 Oct 2026 20:44:03 GMT
Allow: GET, HEAD
Accept-Ranges: bytes
Age: 11815
cf-cache-status: HIT
CF-RAY: a4645340cb75088d-KHI
alt-svc: h3=":443"; ma=86400

HTTP/2 200
date: Tue, 06 Oct 2026 11:28:54 GMT
content-type: text/html; charset=utf-8
server: cloudflare
last-modified: Fri, 02 Oct 2026 16:11:02 GMT
allow: GET, HEAD
accept-ranges: bytes
age: 11805
cf-cache-status: HIT
cf-ray: a46453419e4ac8fc-KHI
alt-svc: h3=":443"; ma=86400
```

"Final rules page showing both LAB2 block rules disabled (not deleted), and Ubuntu client output confirming restored connectivity: ping to 1.1.1.1 (4/4 packets, 0% loss), HTTP 200 OK, and HTTPS 200 OK. The laboratory environment has been successfully restored to its baseline state for the next activity."